

Project Executable Files

SMARTBRIDGE - Project Executable Files

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-6970
- **Team Size:** 3
- **Team Leader:** Prince Kumar P
- **Team Members:** Gowtham L, Guru Moorthi M
- **Project Name:** AI FitTrack
- **Maximum Marks:** 3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)	Remarks / Details
1	Complete source code (all files and folders)	Yes	Full Node.js/Express.js backend source code included.
2	README / Setup Guide (instructions to run)	Yes	Complete README.md file located in the root directory.
3	requirements.txt / package.json / dependency file	Yes	package.json & package-lock.json included.
4	Database schema / seed files (if applicable)	Yes	Mongoose Schema files in src/models/User.js & src/models/Workout.js .
5	Environment configuration file (.env.example)	Yes	.env.example and configured .env included.

6	Deployed application URL / Executable binary	Yes	Local REST API server at http://localhost:5000 .
7	Dockerfile / Containerization config (if applicable)	NA	Standard Node.js runtime environment used.
8	APK (if applicable)	NA	Backend RESTful API service.
9	Test files and test results	Yes	Postman Collection (FitTrack.postman_collection.json) & test logs.
10	Demo video or walkthrough	Yes	Google Drive walkthrough & API demo video links provided.

Step 2: File / Folder Structure

Plaintext

Code Files /

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├── node_modules/
├── src/
│   ├── config/
│   │   └── db.js          # MongoDB connection configuration using Mongoose ODM
│   ├── controllers/
│   │   ├── aiController.js  # Handles Gemini AI recommendations and insights logic
│   │   ├── authController.js # User registration, JWT login, and profile controllers
│   │   └── workoutController.js # CRUD operations for workout management
│   ├── middleware/
│   │   ├── auth.js          # Express JWT authentication & token verification
│   │   └── errorHandler.js  # Centralized error response handling middleware
│   ├── models/
│   │   ├── User.js          # Mongoose schema for User collection (with bcrypt)
│   │   └── Workout.js       # Mongoose schema for Workout collection
│   ├── routes/
│   │   ├── aiRoutes.js      # Express router endpoints for AI services
│   │   ├── authRoutes.js    # Express router endpoints for authentication
│   │   └── index.js         # Central route registry

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├── workoutRoutes.js    # Express router endpoints for workout CRUD
├── services/
│   ├── geminiService.js    # Google Gemini API (@google/genai) service handler
│   ├── app.js              # Express application setup, CORS, and routing middleware
│   └── server.js           # Server startup entry point and DB initialization
├── .env                  # Local environment variables (PORT, MONGO_URI,
GEMINI_API_KEY)
├── .env.example          # Environment variable blueprint
├── .gitignore            # Git ignore file for node_modules and .env
├── FitTrack.postman_collection.json # API collection for Thunder Client / Postman testing
├── package-lock.json     # Locked dependency versions
├── package.json          # Project dependencies and npm startup scripts
└── README.md             # Setup instructions, endpoints guide, and run commands

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Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://localhost:5000/api
Login Credentials (Demo)	Email: sathya@gmail.com Password: Sathya@01
Platform / Hosting Provider	Node.js Express Runtime & MongoDB Atlas Cloud Database
Repository Link	[https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y] (https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y)

Demo Video Link	Demo Video: [https://drive.google.com/file/d/1KswrctSaPI0eOYyJ3GkbbVHvHsU3EfZJ/view?usp=drive_link]https://drive.google.com/file/d/1KswrctSaPI0eOYyJ3GkbbVHvHsU3EfZJ/view?usp=drive_link API Testing Video: [https://drive.google.com/file/d/1TCXG48a-78ryxF3gipdRMwRBwuhAgPaa/view?usp=drive_link]https://drive.google.com/file/d/1TCXG48a-78ryxF3gipdRMwRBwuhAgPaa/view?usp=drive_link
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Step 4: Run Instructions

Pre-requisites:

- **Node.js:** v16.0.0 or higher
- **npm:** v8.0.0 or higher
- **Database:** Active MongoDB Atlas Cloud Cluster or Local MongoDB
- **API Key:** Google Gemini API Key ([@google/genai](#))

Execution Steps:

- **Step 1:** Open VS Code terminal and navigate into project directory:
- PowerShell

cd "Code Files"

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- **Step 2:** Install dependencies:
- PowerShell

npm install

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- **Step 3:** Configure environment variables in `.env` file:
- Code snippet

PORT=5000

MONGO_URI=mongodb+srv://kprince1720_db_user:Princekumar2005@cluster0.6160cws.

mongodb.net/fittrack?retryWrites=true&w=majority

JWT_SECRET=super_secret_fittrack_jwt_key_12345

GEMINI_API_KEY=YOUR_GEMINI_API_KEY

GEMINI_MODEL=gemini-2.5-flash

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- **Step 4:** Start the REST API application server:
- PowerShell

npm start

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- **Step 5:** Confirm running server status in console:
- Plaintext

Server running in development mode on port 5000

MongoDB Connected: cluster0-shard-00-00.6160cws.mongodb.net

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- **Step 6:** Open Thunder Client or Postman and test endpoints at <http://localhost:5000>.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation PDF	Workaround / Status PDF
1	ENOTFOUND MongoDB Connection Delay: Occurs if non-SRV connection strings are used or during internet drops.	Resolved: Upgraded to standard mongodb+srv:// URI format with Mongoose connection retry options.
2	Port Occupied (EADDRINUSE on Port 5000): Port 5000 remains occupied by an orphaned process.	Resolved: Executed npx kill-port 5000 before running npm start .
3	Gemini AI Deprecated Model Error (NOT_FOUND): Older model strings fail API calls.	Resolved: Updated model string to gemini-2.5-flash in .env and geminiService.js .